Analytical Cume

September 7, 2007

Theoretical electrochemical cell potentials and a function of solution concentrations

A solution is prepared by mixing 25.0 mL of 0.20 M KI with 25 mL of 0.200 M NaCl, was titrated with 0.100 M $AgNO_3$.

The experimental arrangement was given by the following cell notation:

S.C.E. || titration solution | Ag

1) Write an expression showing how the cell voltage depends on $[Ag^+]$

2) Calculate the potential of the cell when 25.0 mL of titrant has been added.

3) Calculate the potential of the cell when 50.0 mL of titrant has been added.

4) Calculate the potential of the cell when 75.0 mL of titrant has been added.

5) Calculate the potential of the cell when 100.0 mL of titrant has been added.

6) How much current is drawn through the cell, zero, negligible, a moderate amount, or the amount needed to covert all the silver ion to sliver metal?

Useful Data:

Ksp AgCl is 9.74, Ksp for AgI is 16.08

$Ag^+ + 1e^- \longrightarrow Ag$ $\quad E^o = 0.799$ V

E = 0.241 V for the S.C.E.